

1. System Requirements and Access

- **Device Compatibility:** MouseFrame is optimized exclusively for desktop and laptop displays. It is not supported on mobile devices or tablets due to the precision required for spatial annotation.
- **Access Modes:** Users can ask for a registered account to save calibration data for longitudinal studies or use the Guest Login to trial the application's core features.

2. Image Preparation and Digitization

- **Resolution:** Always scan footprint sheets at a consistent DPI to ensure uniform image resolution across your dataset.
- **Spatial Anchor:** Draw a visible 1 cm reference line on at least one sheet per batch before scanning.
- **Hardware Changes:** If you switch scanners, you must recalibrate the pixel-to-centimetre ratio using a new 1 cm reference line to account for hardware differences.

3. Image Upload and Sizing

- **Workspace:** Navigate to the **Analysis** page to upload compatible footprint images. Upon upload, an accordion-style menu with four main sections will activate.
- **Resizing:** Adjust the image size for comfortable viewing using the interface slider. Alternatively, hold the **Z** key and press + or - on your keyboard.
- **Consistency:** Select an ideal working zoom level and keep it consistent. The image scaling percentage is recorded automatically, as it directly affects spatial calibration.

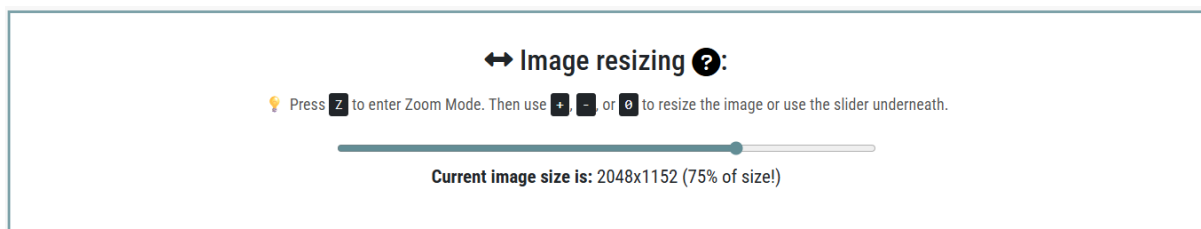


Figure 1: Image resize section. For more information users can click the (?) info button that will display more information on image resizing functionalities and options

4. Point Annotation

Mark the paw placements on the digitized footprint sheet.

- **Point Sizing:** Select your preferred point size before beginning annotation to maintain visual consistency throughout the session.
- **Activation:** Click the corresponding paw on the mouse schematic (e.g., "Red front-left") or use the designated keyboard shortcuts displayed below the schematic. An alert (e.g., "Red (front) Left circle active") will confirm your selection.
- **Correction Tools:** Replace individual points as you work by selecting them from the active point list.
- **Deletion:** Remove specific points from the dynamically updated list, or delete entire groups at once (e.g., "Delete Red L group").

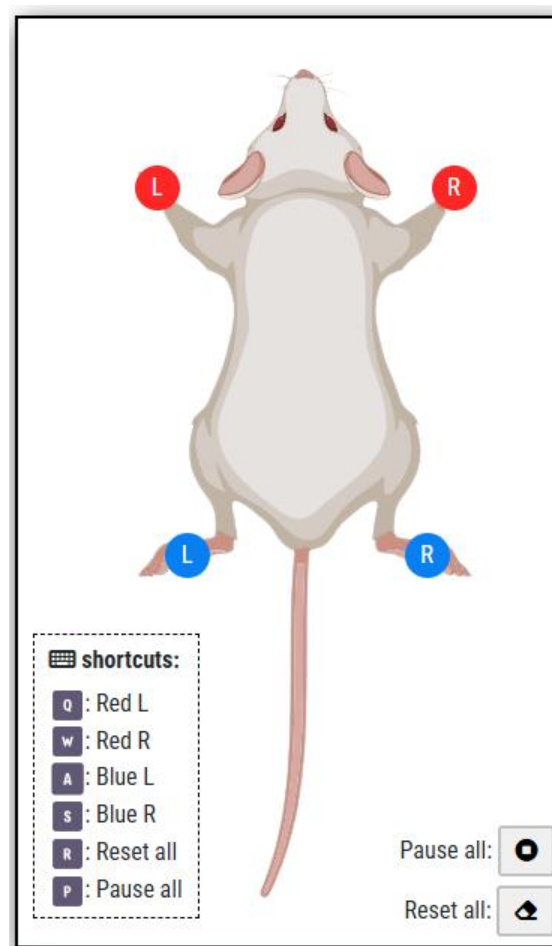


Figure 2: Button for paws in a mouse schematic. In this image there are four buttons that can be clicked to activate the marking of the respective paws, optionally users can use the shortcut keys as shown in the shortcuts' legend.

5. Spatial Calibration

Define how many pixels represent one physical centimetre on your uploaded image.

- **Measurement Tool:** Click the + button to activate the calibration tool. Carefully mark the exact start and end points of the 1 cm reference line on your image.
- **Calculate:** Click the **Ruler** button. The exact pixel distance will be calculated and displayed under "Measurement result."
- **Manual Entry:** If you already know your calibration ratio for a specific image scale, input the value directly into the provided field.

Measure how many pixels in a centimeter

Define how many pixels are in a centimeter:

1. Measuring the 1cm (optional according to availability)

- If there is a drawn line with the length of 1cm, measure it in pixels here by clicking "+" below.
- The "+" button activates the function to add two new points to the image above, delimiting the 1cm line in it.
- After adding to points delimiting the 1cm line, click the ruler icon to measure that distance in pixels.
- The result will appear below.

+

↶

↷

2. Manual input a value or use previous saved values

Manually input known value of pixels in 1 cm or use saved values (choose only one of the options and click use):

Px in cm... OR Select a value Use ↺

Measurement result

Value: px equals 1 cm.

Save section

If you have a value you want to save, use the fields below to add the value in pixels and the size of the image you took the value from (both fields are required to save).

▲ Important: Saving both these values allows for better practice and prevents the usage of such values in different sized images which is not correct.

1. Value(px): Save value...

2. Image size in which the value is being saved: E.g. 1920x1080

Figure 3: Measure how many pixels in a centimetre section: Above are the options to mark points using “+” (in a red square) at the start and end of the 1 cm line in the digitized figure. For measuring the distance between these points click the ruler icon (green square). To stop the function users can click the eraser (orange arrow pointing to it) or click general marking buttons shown before (either pause button – P or reset button – R). After measurement the measurement result will be populated with the value in pixels. Lastly, the save section displays two input boxes for users to save known values of pixels for specific image sizes that need to be saved for future reference.

6. Data Extraction and Management

The Data Management module features a configuration panel on the right and an output display on the left.

- **Configuration:** Input a table title (typically the animal identifier or experimental stage) and specify the left-to-right sequence of the initial footstep pair.

- **Extraction Options:** Click **Add entries** to extract raw pixel spatial relationships (no calibration needed). Click **Add entries with average** to compute precise physical measurements in centimetres (requires prior spatial calibration).
- **Data Table:** The output details entry numbers, evaluated points, measurement types, values (pixels/cm), and the applied image scaling percentage.
- **Curation:** Manually delete individual entries by row number, remove specific measurement types, or clear the entire table using the tools in the right-hand panel.
- **Export:** Export the finalized dataset directly to a Microsoft Excel file or copy it to your system clipboard for external integration.

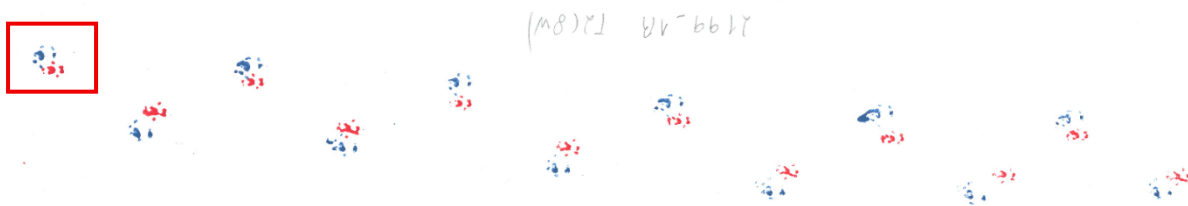


Figure 4: Example of a digitized inked paw test sheet. In this sheet the first pair (red square) of footsteps that appear are on the left side therefore in the data management section users would need to choose “Left”.

Data results Table

Table ID: test				
Entry Number	Points	Measurement type	Value(px) at 33%	Value(cm) at 33%
1	Red L1 → Red L2	Stride length left front	388.021	3.155
2	Red L2 → Red L3	Stride length left front	351.013	2.854
3	Red L3 → Red L4	Stride length left front	412.044	3.350
Stride length left front average in cm:				3.119
5	Red R1 → Red R2	Stride length right front	482.017	3.919
6	Red R2 → Red R3	Stride length right front	302.424	2.459
7	Red R3 → Red R4	Stride length right front	326.345	2.653
Stride length right front average in cm:				3.010
9	Blue L1 → Blue L2	Stride length left hind	361.500	2.939
10	Blue L2 → Blue L3	Stride length left hind	320.141	2.603
11	Blue L3 → Blue L4	Stride length left hind	247.808	2.015

Figure 5: Data results table example. After clicking the “Add entries with average” button and no alerts are displayed, a table such as this will appear. Entry number simply represents the data row index, points column shows between what points the distance is displayed in that row, measurement type presents which type of measurement is in that data row and finally there are values in both pixels and centimetres with the annotation that the user was working with an image resized to approximately 33% of the original.

7. Account Storage and Saved Values

- **Saving Calibration:** Registered users can navigate to the **Save section** under spatial calibration to link their calculated pixel-to-centimetre value with the specific image scale used.
- **Quick Access:** Saved values are tied to the user account and appear instantly in the "Manually input a value" dropdown menu for future sessions.
- **Guest Limitations:** Guest accounts cannot save calibration values to the database and must calibrate manually per session.

External Resources

- **Application Website:** <https://mouseframe.pt/>
- **Support:** mouseframe.app@gmail.com, bernaz.estevam97@gmail.com or mngt@ualg.pt